

The turkey problem revisited: what dimensional analysis can and cannot decide

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<https://github.com/l0d0v1c/TurkeyDimensional>

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Abstract

A textbook exercise in dimensional analysis asks how long a 10lb turkey takes to roast given that a 5lb turkey takes five hours. Two things are wrong with it. First, its data are fictitious: published roasting charts give roughly three hours for a 10lb bird, not five or eight. Second, the standard answer $t \propto m^{2/3}$ is presented as *the* consequence of dimensional analysis when it is in fact the consequence of one modelling choice among several. We show that enumerating dimensionally admissible monomials over an over-specified variable set — one that contains both the thermal diffusivity α and its constituents k , ρ , c , and both the mass m and the length l they determine — generates sixteen formulae that collapse to six, and that ten of those sixteen retain the absolute temperature T even though the governing equation is linear in temperature and cannot contain it. Correctly posed, the problem yields a single group, $Fo = t\alpha/l^2 = F(Bi, \theta)$. We then fit published roasting charts by weighted log-log regression rather than by correlation coefficients, which we show are incapable of discriminating between candidate exponents. The measured exponent is $n = 0.54 \pm 0.02$, excluding $n = 1$ at 7.5σ and lying below the isothermal-surface value $2/3$. A finite-Biot calculation reproduces $n_{\text{eff}} = 0.54\text{--}0.60$ for realistic oven heat-transfer coefficients, so the deviation is a consequence of the diffusive model, not evidence against it. A corrected version of the exercise is proposed.

Keywords: dimensional analysis; Buckingham π theorem; transient conduction; Biot number; scaling exponents; physics education

1 Introduction

Dimensional analysis [1, 2] is taught partly through short problems whose answer is counter-intuitive. One of the most widely reproduced is the roasting turkey [4, 5]: *a 5lb turkey needs five hours to cook; how long does a 10lb turkey need at the same oven temperature?* The expected answer is not ten hours but $5 \times 2^{2/3} \approx 8$ hours, from $t \propto l^2$ and $l \propto m^{1/3}$.

The exercise is rarely checked against data. When it is, two distinct problems appear.

The first is factual. Published roasting charts [6, 7] give 2.75 h to 3.0 h for an unstuffed 10lb bird. Neither the premise (5lb in 5 h) nor the answer (8 h) corresponds to anything a cook would recognise. A student who checks the result learns that the method gives wrong numbers, which is the opposite of the intended lesson.

The second is methodological, and is our main concern. The $2/3$ law is routinely presented as a *result* of dimensional analysis. It is not. Dimensional analysis alone, applied to a carelessly chosen list of variables, admits many mutually inconsistent power laws; the $2/3$ law follows only once the physics has been used to restrict that list. Failing to make this explicit invites a tempting but invalid

procedure: enumerate all admissible monomials, then let data choose among them. Section 2 shows why that procedure fails, Section 3 gives the correct formulation, and Sections 4–5 extract the exponent that roasting charts actually encode.

2 Enumeration over a redundant variable set

Suppose we do not know which laws govern the problem and list the quantities that might plausibly enter: the time t , mass m , a characteristic half-thickness l , density ρ , specific heat c , thermal conductivity k , thermal diffusivity α , and temperature T . With four independent dimensions (M , L , Θ , and time), any subset of five quantities including t yields exactly one dimensionless group, hence one monomial $t \propto \prod x_i^{a_i}$. Enumerating over subsets produces sixteen such formulae, apparently offering mass exponents 1, 2/3, 1/2, 4/9, 2/5 and 1/3 and leaving the choice open.

The abundance is an artefact, for two reasons.

2.1 The list is not independent

Thermal diffusivity is not a new quantity but a combination, $\alpha = k/(\rho c)$, and for a body of fixed shape and density the mass and the half-thickness are not independent either, $m = \kappa \rho l^3$ with κ a shape factor. Substituting both identities collapses the sixteen formulae onto six distinct relations, listed in Table 1.

Table 1: The sixteen admissible monomials, reduced on the independent set (ρ, l, c, T, α) . Labels index the sixteen monomials in the order produced by subset enumeration; the explicit list is generated by the accompanying script.

Monomials	Reduced relation	Contains T ?
1, 2, 6, 7, 11, 12	$t \propto l^2/\alpha$	no
4, 8, 9	$t \propto \alpha/(cT)$	yes
5, 13, 3	$t \propto l^{4/3} (cT\alpha)^{-1/3}$	yes
10, 14	$t \propto l (cT)^{-1/2}$	yes
15	$t \propto l^{3/2} c^{-1/4} T^{-1/4} \alpha^{-1/2}$	yes
16	$t \propto l^{6/5} (cT)^{-2/5} \alpha^{-1/5}$	yes

Six of the sixteen — including the one that appears to give $t \propto m$, namely $t \propto mc/(lk)$ — reduce to $t \propto l^2/\alpha$, the classical result. Their apparent disagreement is purely notational: $t \propto mc/(lk)$ differs from $t \propto \rho l^2 c/k$ only by the substitution $m = \kappa \rho l^3$. Treating them as rival hypotheses, and in particular varying m at fixed l , is not permitted: at fixed density and shape those two variables move together by definition.

2.2 Absolute temperature cannot appear

Ten of the sixteen retain T . This too is excluded, and on grounds that are available before any data are collected. The heat equation is linear in temperature. Writing $\theta = (T_{\text{oven}} - T)/(T_{\text{oven}} - T_{\text{init}})$, the temperature scale cancels identically from both the equation and its boundary conditions, and the doneness criterion is itself a condition on θ — the core must reach 74°C in an oven at 163°C from a refrigerated start, i.e. $\theta_c \simeq 0.56$. No absolute temperature survives.

One may check this independently on dimensional grounds. Of the listed quantities, only T , c and k carry Θ , and Θ can be cleared only through the products cT and kT . But $[cT] = L^2 \text{time}^{-2}$,

a squared velocity: every relation in Table 1 that contains T is an acoustic or energy-per-mass scaling, not a diffusive one. Such formulae predict, for instance, that doubling the absolute oven temperature shortens cooking by a factor $2^{1/4}$ regardless of the target core temperature, which is plainly false — an oven at 74°C never finishes at all.

The lesson is not that dimensional analysis is unreliable but that it is exactly as good as the variable list it is given. Enumerating monomials over a redundant list does not reveal hidden alternatives; it manufactures them.

3 Correct formulation

The physically admissible statement involves two dimensionless parameters, not one power law. For transient conduction into a body of half-thickness l with surface coefficient h ,

$$Fo = \frac{t\alpha}{l^2} = F(Bi, \theta_c), \quad Bi = \frac{hl}{k}, \quad (1)$$

which is what the π theorem actually delivers once the list is clean. Equation (1) is not a monomial, and the widely quoted $t \propto m^{2/3}$ is its $Bi \rightarrow \infty$ limit at fixed θ_c , where F becomes a constant.

At finite Bi the exponent is no longer $2/3$, because Bi itself grows with size. Using the leading-term solution for a sphere, $\theta = A_1(Bi) \exp[-\lambda_1^2(Bi)Fo]$ with $\lambda_1 \cot \lambda_1 = 1 - Bi$ [3], one has $Fo = \lambda_1^{-2} \ln(A_1/\theta_c)$ and therefore

$$n_{\text{eff}} \equiv \frac{d \ln t}{d \ln m} = \frac{1}{3} \left(2 + \frac{d \ln Fo}{d \ln Bi} \right) < \frac{2}{3}, \quad (2)$$

since Fo decreases with Bi . Taking $\rho = 1050 \text{ kg m}^{-3}$, $c = 3200 \text{ J kg}^{-1} \text{ K}^{-1}$, $k = 0.45 \text{ W m}^{-1} \text{ K}^{-1}$ (hence $\alpha = 1.34 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$) and $h = 10 \text{ W m}^{-2} \text{ K}^{-1}$ to $25 \text{ W m}^{-2} \text{ K}^{-1}$ for a domestic oven, birds of 6 lb to 28 lb span $Bi \approx 2$ –8, and Eq. (2) gives $n_{\text{eff}} = 0.54$ – 0.60 (Fig. 1c). A measured exponent below $2/3$ is thus predicted by the diffusive model, not evidence against it.

4 Data and fitting method

4.1 Why correlation coefficients must not be used

A natural temptation is to test candidate exponents x by computing the correlation $r(m^x, t)$ and keeping the largest. This does not work. Over a mass range of a factor five, $m^{1/2}$ and $m^{2/3}$ are both smooth increasing and nearly affine functions of one another, so r is close to unity for every candidate. On the chart of Table 2, r varies by less than 0.004 across the whole interval $x \in [0.3, 1.0]$ and peaks near $x \simeq 0.75$ (Fig. 1b). Differences in the third decimal of r carry no information about x , and rankings built on them are not reproducible.

4.2 Regression

We instead regress $\ln t$ on $\ln m$ directly, so that the exponent is the fitted slope and comes with an interval. Chart entries are published as ranges in both variables; we take geometric midpoints as central values and half-widths in log space as uncertainties, and weight accordingly. Data are given in Table 2.

Two caveats apply to these data and limit what can be asked of them. The published intervals overlap in mass, so a 12 lb bird appears in two rows with different times; and chart times are conservative by design, carrying a food-safety margin rather than reporting a measurement. They constrain a scaling exponent to within a few percent, not better.

Table 2: Published roasting times, oven at 325 °F (163 °C), to a core temperature of 165 °F (74 °C) [7]. Note that 165 °F is the doneness endpoint, not the oven setting — a confusion present in earlier treatments.

Mass (lb)	Unstuffed (h)	Stuffed (h)
6–8	2.25–2.75	2.75–3.00
8–12	2.75–3.00	3.00–3.50
12–14	3.00–3.75	3.50–4.00
14–18	3.75–4.25	4.00–4.25
18–20	4.25–4.50	4.25–4.75
20–24	4.50–5.00	4.75–5.25
24–28	5.00–5.50	5.25–5.75

5 Results

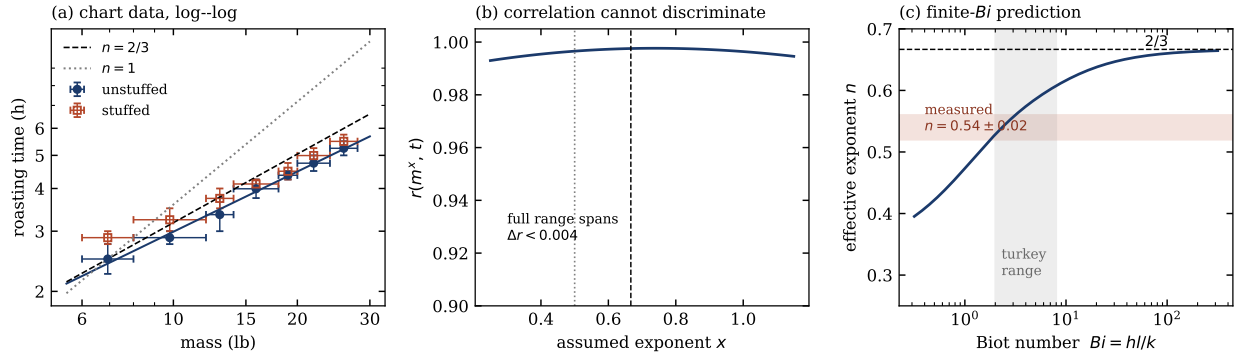


Figure 1: (a) Chart data with fitted power law; the $n = 1$ and $n = 2/3$ lines are anchored at the smallest bird. (b) Correlation coefficient $r(m^x, t)$ against assumed exponent x : the whole range spans less than 0.004, so r cannot discriminate. (c) Effective exponent from Eq. (2) against Biot number, with the range spanned by 6 lb to 28 lb birds shaded and the measured value in red.

Weighted log-log regression gives

$$\text{unstuffed: } n = 0.61 \pm 0.05, \quad (3)$$

$$\text{stuffed: } n = 0.49 \pm 0.04, \quad (4)$$

$$\text{joint fit (common exponent): } n = 0.54 \pm 0.02, \quad (5)$$

the joint fit allowing separate prefactors, which differ by a factor 1.08. Referred to the unstuffed fit, the proportional law $n = 1$ is excluded at 7.5σ ; $n = 2/3$ lies at 1.1σ and $n = 1/2$ at 2.0σ . The fitted relation is

$$t [\text{h}] = 0.78 (m [\text{lb}])^{0.59}, \quad (6)$$

valid over 6 lb to 28 lb at 325 °F.

Three points deserve emphasis. First, the two columns do not return the same exponent, which sets a floor on what these data can resolve: the honest statement is $n \approx 0.55 \pm 0.05$, not a value to three decimals. Second, that value is consistent with the finite-Biot prediction of Section 3, so no departure from diffusive physics need be invoked. Third, a competing explanation exists and

cannot be separated here: a turkey is not self-similar, and large birds grow in length and breadth more than in thickness, so the relevant half-thickness scales as $l \propto m^p$ with $p < 1/3$, depressing n by the same amount. Distinguishing the two mechanisms requires thickness measurements, not chart entries.

6 A second data set: bread, and why it does not help

Baking charts [8] are sometimes offered as an independent check. They are not usable for this purpose, and the reason is instructive. In the chart we examined, oven temperature falls as loaf mass rises (523 K for small rolls, 488 K for a 1 kg boule): $\ln m$ and $\ln T$ have a sample correlation of -0.75 . Mass and temperature effects are therefore collinear and not separately identifiable. A mass-only fit returns $n = 0.53$ with $R^2 = 0.89$; adding a free temperature exponent raises R^2 to 0.99 but returns an exponent of -8.8 on T , which is physically meaningless and simply absorbs the collinearity. The improvement is a property of the design matrix, not evidence for a temperature correction.

Bread differs from poultry in other ways that matter: a loaf loses 15 % to 20 % of its mass to evaporation during baking, and its doneness criterion involves crust formation rather than a core temperature alone. We report the analysis only to caution against it.

7 A corrected exercise

The pedagogical value of the problem survives if it is posed on real numbers and with the modelling step made explicit. We suggest:

A 24 lb turkey roasts in 5.2 h at 325 °F. (a) Assuming the bird is geometrically similar and that heat transport is diffusion-limited with an isothermal surface, estimate the time for a 10 lb bird. (b) The published chart gives 2.75 h to 3.0 h. Your estimate is slightly high. Identify which assumption fails, and show that relaxing it moves the prediction in the right direction.

Part (a) gives $5.2 \times (10/24)^{2/3} = 2.9$ h, inside the chart range. Part (b) is the point of the exercise: the isothermal-surface assumption is $Bi \rightarrow \infty$, whereas $Bi \approx 3$ here, and Eq. (2) lowers the exponent towards the measured 0.55. Students end with a correct number, an explicit model, and a named assumption to question — rather than with eight hours and a turkey that would be inedible.

8 Conclusion

The turkey problem is a good exercise built on bad data and an unstated modelling step. Three conclusions follow.

Dimensional analysis does not select among power laws; the variable list does. Enumerating admissible monomials over a list containing both α and k, ρ, c , and both m and l , produced sixteen formulae that reduce to six, ten of them containing an absolute temperature that the linearity of the heat equation forbids. Correctly posed, the problem gives one group and two parameters, $Fo = F(Bi, \theta_c)$.

Correlation coefficients cannot be used to choose an exponent. On these data they vary by less than 0.004 over $x \in [0.3, 1.0]$ and peak at the wrong value. Log-log regression with stated uncertainties is the minimum acceptable tool.

The measured exponent, $n = 0.54 \pm 0.02$, is below $2/3$ and far from 1. Both a finite Biot number and departure from geometric similarity predict a deficit of the observed size, and chart data cannot separate them. Resolving that is the natural experiment: instrument birds of several masses with core probes, record thickness as well as mass, and repeat at two oven temperatures so that θ_c is varied independently.

Data and code availability

All fits, the monomial reduction of Table 1, and Fig. 1 are reproducible from the scripts accompanying this note, available at <https://github.com/l0d0v1c/TurkeyDimensional>. *[Add repository DOI before submission.]*

Declaration of competing interest

[If any software cited here is authored or distributed by the authors, declare it explicitly in this section.]

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- [8] Baking time chart for bread. *[Add publisher and access date.]*